

CHAPTER 1: STRUCTURE AND BONDING

1.1 Carbon Chemistry

Carbon has 4 valence electrons and forms 4 covalent bonds. This gives carbon an extraordinary ability to form diverse molecular structures: chains, branches, and rings.

1.2 Hybridization

sp³ hybridization: 4 sigma bonds, tetrahedral (109.5 degrees)

Example: methane CH₄, ethane C₂H₆

sp² hybridization: 3 sigma + 1 pi bond, trigonal planar (120 degrees). Example: ethene C₂H₄, benzene C₆H₆

sp hybridization: 2 sigma + 2 pi bonds, linear (180 degrees)

Example: ethyne C₂H₂ (acetylene)

1.3 Bond Polarity

Electronegativity increases across a period and decreases down a group. Bond polarity arises from differences in electronegativity between bonded atoms.

Common bond polarities (most to least polar):

O-H > N-H > C-H > C-C

CHAPTER 2: HYDROCARBONS

2.1 Alkanes (C_nH_{2n+2})

Saturated hydrocarbons with only single bonds.

Named with prefix based on carbon count:

1C: methane 5C: pentane 9C: nonane

2C: ethane 6C: hexane 10C: decane

3C: propane 7C: heptane

4C: butane 8C: octane

Properties:

- Nonpolar, insoluble in water
- Low melting/boiling points

- Combust readily (fuel sources)
- Unreactive (only undergo substitution at high T/P)

2.2 Alkenes (C_nH_{2n})

Unsaturated hydrocarbons with at least one $C=C$ double bond.

Named with '-ene' suffix.

ethene (ethylene): $H_2C=CH_2$

propene (propylene): $CH_3-CH=CH_2$

Key Reactions:

- Hydrogenation: add H_2 across double bond (with Pt/Pd catalyst)
- Halogenation: add X_2 (Br_2 , Cl_2) across double bond
- Hydrohalogenation: add HX (Markovnikov addition)
- Hydration: add H_2O (acid-catalyzed, Markovnikov)

2.3 Alkynes (C_nH_{2n-2})

Contain at least one $C-C$ triple bond. Named with '-yne' suffix.

ethyne (acetylene): $HC\equiv CH$

CHAPTER 3: FUNCTIONAL GROUPS

Functional groups determine the chemical properties of organic molecules. They are specific arrangements of atoms that react in predictable ways.

3.1 Hydroxyl Group (-OH) - Alcohols

General formula: $R-OH$

Classification: primary (1 degree), secondary (2), tertiary (3)
based on how many carbons the -OH carbon is attached to.

Properties:

- Polar (can form hydrogen bonds)
- Higher boiling points than equivalent alkanes
- Soluble in water (for small molecules)

Key Reactions:

- Dehydration: alcohol $\rightarrow$ alkene (with H_2SO_4 , heat)
- Oxidation: primary alcohol $\rightarrow$ aldehyde $\rightarrow$ carboxylic acid
secondary alcohol $\rightarrow$ ketone

tertiary alcohol -> no reaction

3.2 Carbonyl Group (C=O)

Aldehydes (R-CHO): carbonyl at end of chain

Formaldehyde: H-CHO

Acetaldehyde: CH₃-CHO

Ketones (R-CO-R'): carbonyl in middle of chain

Acetone: CH₃-CO-CH₃

Both have polar C=O bond, but aldehydes more reactive than ketones due to less steric hindrance.

3.3 Carboxyl Group (-COOH) - Carboxylic Acids

General formula: R-COOH

Weak acids (donate H⁺ from -OH group)

Examples:

Formic acid: HCOOH (found in ant stings)

Acetic acid: CH₃COOH (vinegar)

Citric acid: found in citrus fruits

pK_a values indicate acid strength:

Formic acid: 3.75

Acetic acid: 4.76

(Lower pK_a = stronger acid)

3.4 Amino Group (-NH₂) - Amines

General formula: R-NH₂ (primary amine)

R₂-NH (secondary)

R₃-N (tertiary)

Properties:

- Basic (accept H⁺ from water)
- Good nucleophiles
- Found in amino acids, proteins, DNA bases

CHAPTER 4: STEREOCHEMISTRY

4.1 Isomers

Structural Isomers: same formula, different connectivity

Example: butane and isobutane (both C_4H_{10})

Stereoisomers: same connectivity, different spatial arrangement

- cis/trans (geometric): around $C=C$ or ring
- enantiomers: non-superimposable mirror images

4.2 Chirality

A molecule is chiral if it is not superimposable on its mirror image. The most common cause is a carbon with four different substituents (chiral center or stereocenter).

Enantiomers have identical physical properties but rotate plane-polarized light in opposite directions.

R/S naming: based on priority of substituents (CIP rules)